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Terminal device, network node, and control method for communication control with consideration given to antenna arrangement

Abstract

A terminal device that comprises a plurality of antennas notifies a network to which the terminal device belongs of information according to which it is possible to specify at least one of, among the plurality of antennas, a second antenna having a predetermined characteristic in common with a first antenna included in the plurality of antennas, and a third antenna not having the predetermined characteristic in common with the first antenna.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation of International Patent Application No. PCT/JP2023/006707 filed on Feb. 24, 2023, which claims priority to and the benefit of Japanese Patent Application No. 2022-053611, filed on Mar. 29, 2022, the entire disclosures of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a communication control technology according to an arrangement of antennas in a terminal device.

Description of the Related Art

(2) In accordance with the standard of the Third Generation Partnership Project (3GPP), wireless communication systems are widely used in which a plurality of geographically distributed base

station devices provide a wireless communication service to a terminal device within a range of electromagnetic waves transmitted by each base station device. In such a wireless communication system, various types of communication control are performed according to the state of a transmission path between an antenna of the base station device and an antenna of the terminal device.

(3) In response to wireless communication being used in a variety of situations, the configurations of terminal devices are becoming more diverse. For example, not only small terminal devices such as conventional mobile phones, but also terminal devices mounted on large objects such as vehicles can be used. In addition, in fixed wireless access (FWA) using a cellular communication system, a wireless device on the user side can be used as a terminal device of the cellular communication system.

(4) Conventionally, when a terminal device has a plurality of antennas, it is assumed that all of these antennas are arranged inside a small terminal device, and therefore the characteristics of the transmission paths between the plurality of antennas and the antenna of the base station device are generally treated as being the same. On the other hand, as terminal devices become more diverse, it is assumed that the plurality of antennas of the terminal devices are arranged at locations spaced apart from each other, and the characteristics of the transmission paths between the plurality of antennas and the antenna of the base station device will differ significantly. In such a case, if the characteristics of the transmission paths in the plurality of antennas of the terminal device are treated as being generally the same, it is difficult to obtain sufficient communication quality.

SUMMARY OF THE INVENTION

(5) The present invention provides a method for enabling execution of appropriate communication control corresponding to an arrangement of antennas in a terminal device.

(6) A terminal device according to one aspect of the present invention is a terminal device comprising: a plurality of antennas; at least one processor; and at least one memory that stores a computer-readable instruction for causing, when executed by the at least one processor, the at least one processor to: notify a network to which the terminal device belongs of information according to which it is possible to specify at least one of, among the plurality of antennas, a second antenna having a predetermined characteristic in common with a first antenna included in the plurality of antennas, and a third antenna not having the predetermined characteristic in common with the first antenna.

(7) A network node according to one aspect of the present invention is a network node comprising: at least one processor; and at least one memory that stores a computer-readable instruction for causing, when executed by the at least one processor, the at least one processor to: acquire, from a terminal device belonging to a network including the network node, information according to which it is possible to specify at least one of, among a plurality of antennas included in the terminal device, a second antenna having a predetermined characteristic in common with a first antenna included in the plurality of antennas, and a third antenna not having the predetermined characteristic in common with the first antenna; and perform, based on the information, control of communication of the terminal device, the control being the same for the first antenna and the second antenna, and the control being different for the first antenna and the third antenna.

(8) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate embodiments of the invention and, together with the description, serve to

explain principles of the invention.

(2) FIG. 1 is a diagram illustrating an example of a system configuration.

(3) FIG. 2 is a diagram illustrating an example of a hardware configuration of a device.

(4) FIG. 3 is a diagram illustrating an example of a functional configuration of a terminal device.

(5) FIG. 4 is a diagram illustrating an example of a functional configuration of a base station device.

(6) FIG. 5 is a diagram illustrating an example of a flow of processing executed in the system.

(7) FIG. 6 is a diagram illustrating an example of a flow of processing executed in the system.

(8) FIG. 7 is a diagram illustrating an example of a flow of processing executed in the system.

DESCRIPTION OF THE EMBODIMENTS

(9) Hereinafter, embodiments will be described in detail with reference to the attached drawings. Note, the following embodiments are not intended to limit the scope of the claimed invention, and limitation is not made to an invention that requires a combination of all features described in the embodiments. Two or more of the multiple features described in the embodiments may be combined as appropriate. Furthermore, the same reference numerals are given to the same or similar configurations, and redundant description thereof is omitted.

(10) System Configuration

(11) FIG. 1 shows an example of a configuration of a wireless communication system according to this embodiment. The wireless communication system may be, for example, a cellular communication system conforming to the fifth generation (5G) standard standardized by the 3rd Generation Partnership Project (3GPP). However, this is just one example, and the following discussion may also be applied to wireless communication systems conforming to conventional standards such as Long Term Evolution (LTE), standards from the sixth generation and onward, or the like, and may also be applied to wireless communication systems other than cellular communication systems.

(12) This wireless communication system includes, for example, a base station device **101** and a terminal device **111**. Note that the base station device **101** is a wireless access network node connected to a core network, and the network may of course also include other network nodes, but for simplicity of description, such configurations are omitted in FIG. 1. Here, it is assumed that the terminal device **111** has a plurality of (at least two) antennas, and that among the plurality of antennas, an antenna **121** and an antenna **122** are arranged at positions spaced significantly apart from an antenna **123** and an antenna **124**. In this case, it is assumed that the antenna **121** and the antenna **122** have roughly the same characteristics with respect to the electromagnetic waves arriving from the base station device **101**. For example, the reception timings of electromagnetic waves received by the antenna **121** and the antenna **122** are roughly the same, the same settings can be used for downlink reception timing control, such as timing control for Fourier transform, and the same settings can be used for uplink transmission timing control, such as timing advance. That is, in the communication via the antenna **121** and the antenna **122**, common communication timing control (timing control in at least one of the uplink and the downlink) can be used. Similarly, the antenna **123** and the antenna **124** are assumed to have roughly the same characteristics with respect to electromagnetic waves arriving from the base station device **101**, and for example, common communication timing control can be applied to them. On the other hand, it can be assumed that the antenna **121** or antenna **122** and the antenna **123** or antenna **124** have different characteristics from each other. For this reason, if common communication timing control is applied to the antenna **121** and the antenna **123**, it may be impossible to perform communication appropriately. For example, in the downlink, a Fourier transform window setting that is appropriate for the antenna **121** may not be appropriate for the antenna **123**, making it impossible to ensure orthogonality.

(13) Also, for example, the terminal device **111** can use a plurality of panels to communicate with different transmission and reception points within the network. In this case, since it is assumed that

the characteristics will differ significantly from panel to panel, it may not be appropriate to use a common setting for communication using a plurality of panels. Note that although separate settings can be applied to each of the plurality of antennas, the processing can become more complicated as the number of antennas increases.

(14) For this reason, in this embodiment, when electromagnetic waves arriving from the base station device **101** are detected via a plurality of antennas, it is possible to decide whether to apply the same setting control or a different setting control depending on whether or not the respective predetermined characteristics are the same as each other. For this reason, the terminal device **111** notifies the base station device **101** (a network node within the network) of, for example, information that enables the base station device **101** to specify at least one of a second antenna that has a predetermined characteristic in common with a first antenna among the plurality of antennas that are included in the terminal device **111**, and a third antenna that does not have the predetermined characteristic in common. For example, the terminal device **111** can notify the base station device **101** that the antenna **121** and the antenna **122** have a predetermined characteristic in common. In addition, the terminal device **111** may notify the base station device **101** that the antenna **121** does not have the predetermined characteristic in common with the antenna **123** and the antenna **124**. Also, the terminal device **111** may notify the base station device **101** that, for example, the antenna **121** and the antenna **122** belong to a first group, and the antenna **123** and the antenna **124** belong to a second group. In this case, it can be indicated that two antennas in the same group have a predetermined characteristic in common with each other, and two antennas in different groups do not have a predetermined characteristic in common with each other. Based on this notification, the network node (e.g., base station device **101**) performs setting and control of communication that is the same for an antenna group that has the predetermined characteristic in common, and performs setting and control of communication independently for an antenna group that does not have the predetermined characteristic in common.

(15) In one example, the predetermined characteristic relates to the magnitude of delay observed in electromagnetic waves arriving from the base station device **101** and the base station device **102**. In addition, the predetermined characteristic may relate to the magnitude of delay until electromagnetic waves transmitted from the antennas **121** to **124** of the terminal device **111** arrive at the base station device **101** or the base station device **102**. Note that in the following, for example, for a predetermined antenna of the terminal device **111**, the magnitude of delay at that particular antenna is expressed by including the magnitude of delay observed by the terminal device **111** in the downlink and the magnitude of delay observed by the base station device **101** or the base station device **102** in the uplink. In this case, based on the differences between the magnitude of delay at the antenna **121** and the magnitudes of delay at the other antennas, an antenna for which the difference does not exceed a predetermined value is treated as an antenna that has the predetermined characteristic in common with the antenna **121**. In addition, an antenna for which the difference exceeds the predetermined value can be treated as an antenna that does not have the predetermined characteristic in common with the antenna **121**. Note that the magnitude of the delay in this context can be the average value of the time difference between when electromagnetic waves transmitted from the base station device **101** or the base station device **102** arrive at the terminal device **111** for each antenna. In addition, the magnitude of the delay at each antenna may be specified using the timing at which the electromagnetic waves arrive at any one of the antennas as a criterion.

(16) The network node can apply the same communication timing control to a group of antennas in which the timing at which electromagnetic waves transmitted from the base station device **101** or the base station device **102** arrive at the terminal device **111** is sufficiently close, that is, a group of antennas that have the same characteristic regarding the magnitude of delay. On the other hand, if the network node applies the same communication timing control to an antenna group that do not have the characteristic regarding the magnitude of delay in common, it may be difficult to ensure

orthogonality, and therefore the network node can apply independent communication timing controls to these antennas. The communication timing control can include, for example, setting the timing advance used when transmitting an uplink signal. In addition, the network node can notify the terminal device **111** of setting information that enables execution of different reception controls, for example, setting Fourier transform windows corresponding to different timings, for the antennas **121** and **122** and the antennas **123** and **124**. In addition, the network node can perform the same setting and control of uplink transmission power control for an antenna group that has a characteristic related to the magnitude of delay in common, and can independently perform setting and control of uplink transmission power control for an antenna group that has different characteristics related to the magnitude of delay. It is assumed that antennas with significantly different average delays will have different magnitudes of propagation loss due to the difference in the magnitude of the average delay. For this reason, common transmission power control can be set and controlled for an antenna group that has roughly the same average delay, and transmission power control can be set and controlled independently for an antenna group that does not have the same average delay.

(17) In one example, the predetermined characteristic relates to the magnitude of delay spread observed for electromagnetic waves arriving at the base station device **101** or the base station device **102** from each of the antennas **121** to **124** of the terminal device **111**. In this case, based on the difference between the magnitude of the delay spread for the antenna **121** and the magnitude of the delay spread for other antennas, an antenna for which the difference does not exceed a predetermined value is treated as an antenna that has the predetermined characteristic in common with the antenna **121**. In addition, an antenna for which the difference exceeds the predetermined value can be treated as an antenna that does not have the predetermined characteristic in common with the antenna **121**. In addition, when a correlation value between the shape of the delay spread for the antenna **121** and the shape of the delay spread is calculated, an antenna for which the correlation value exceeds a predetermined value may be treated as an antenna that has the predetermined characteristic in common with the antenna **121**. In addition, an antenna for which the correlation value does not exceed a predetermined value can be treated as an antenna that does not have the predetermined characteristic in common with the antenna **121**.

(18) A network node can give, for example, the same Sounding Reference Signal (SRS) setting to an antenna group that has the delay spread characteristic in common. On the other hand, the SRS setting can be applied independently to an antenna group that does not have the delay spread characteristic in common. Here, the SRS setting can include, for example, setting of the density of frequency resources over which the SRS is to be transmitted. That is, the number of subcarriers on which the SRS is transmitted can be set to be different for each of the plurality of antennas of the terminal device **111**. In this way, a common SRS setting is applied to an antenna group that has the delay spread characteristic in common, and separate settings are applied independently to an antenna group that does not have the delay spread characteristic in common, whereby an SRS setting appropriate for the transmission path can be used for each of the plurality of antennas with significantly different transmission path conditions.

(19) In addition, in one example, the predetermined characteristic relates to the direction in which the electromagnetic waves transmitted from the base station device **101** or the base station device **102** arrive at the terminal device **111**. In this case, based on the difference between the direction of arrival of electromagnetic waves at the antenna **121** and the direction of arrival of electromagnetic waves at other antennas, an antenna for which the difference does not exceed a predetermined value is treated as an antenna that has the predetermined characteristic in common with the antenna **121**. In addition, an antenna for which the difference exceeds the predetermined value can be treated as an antenna that does not have the predetermined characteristic in common with the antenna **121**. The network node can use the same settings and controls for beam control for an antenna group that has the characteristic of the direction of arrival of electromagnetic waves in

common. On the other hand, settings and controls for beam control can be applied independently to an antenna group that does not have the characteristic of the direction of arrival of electromagnetic waves in common. Here, the settings for beam control may include, for example, a channel state information-reference signal (CSI-RS) setting.

(20) Note that the above-mentioned predetermined characteristic can be specified in a fixed manner depending on the installation positions of the antennas **121** to **124** in the terminal device **111**. For example, as shown in FIG. 1, if the antenna **121** and the antenna **122** are located sufficiently close to each other and are located at different positions from the antenna **123** and the antenna **124**, the antenna **121** and the antenna **122** can be treated as having the predetermined characteristic in common. Similarly, the antenna **123** and the antenna **124** can be treated as having the predetermined characteristic in common. On the other hand, the antenna **121**, the antenna **123**, and the antenna **124** can be treated as not having the predetermined characteristic in common, and the antenna **122**, the antenna **123**, and the antenna **124** can also be treated as not having the predetermined characteristic in common. Note that the terminal device **111** may notify the network of information indicating which characteristics are held in common or not among the above-mentioned predetermined characteristics, together with the above-mentioned information. Note that this notification can be given as capability information (UE Capability), for example. In this case, the terminal device **111** can perform notification of the above-mentioned capabilities by, for example, transmitting a message (UECapabilityInformation) performing notification of the capability information to the base station device **101** in response to a capability information inquiry message (UECapabilityEnquiry) from the base station device **101**. In addition, the base station device **101** can set, for example, a radio resource control (RRC) layer based on the acquired information, and can also execute setting and control of the terminal device **111** using, for example, downlink control information (DCI) or the like.

(21) Device Configuration

(22) Next, the configurations of the base station device **101** and the terminal device **111** that execute the above-mentioned processing will be described. FIG. 2 is a diagram illustrating an example of the hardware configuration of the base station device **101** and the terminal device **111**. In one example, the base station device **101** and the terminal device **111** include a processor **201**, a ROM **202**, a RAM **203**, a storage device **204**, and a communication circuit **205**. The processor **201** is a computer including one or more processing circuits, such as a general-purpose CPU (Central Processing Unit) or an ASIC (Application Specific Integrated Circuit), and executes the overall processing of the device and each of the above-mentioned processes by reading out and executing programs stored in the ROM **202** or the storage device **204**. The ROM **202** is a read-only memory that stores information such as programs related to the processes executed by the base station device **101** and the terminal device **111**, and various parameters. The RAM **203** is a random access memory that functions as a workspace when the processor **201** executes a program, and the RAM **203** stores temporary information. The storage device **204** is constituted by, for example, a removable external storage device or the like. The communication circuit **205** includes, for example, a circuit for communicating with another device. Note that although FIG. 2 illustrates one communication circuit **205**, the base station device **101** and the terminal device **111** can have a plurality of communication circuits. Note that the terminal device **111** of this embodiment may include a plurality of antennas, the plurality of antennas may be connected to one communication circuit **205**, or one or more of the plurality of antennas may be connected to one communication circuit **205** and the other antennas may be connected to another communication circuit **205**.

(23) FIG. 3 is a diagram showing an example of a functional configuration of the terminal device **111**. The terminal device **111** includes, for example, an information notification unit **301** and a processing unit **302**. The functions illustrated in FIG. 3 can be realized, for example, by the processor **201** executing a program stored in the ROM **202** or the storage device **204**. Note that this is just one example, and some or all of the functions shown in FIG. 3 may be realized by dedicated

hardware. Also, FIG. 3 shows only the functions related to this embodiment among the functions of the terminal device **111**, and the terminal device **111** can of course have functions other than those shown in FIG. 3, such as functions naturally included in a terminal device in a cellular communication system.

(24) The information notification unit **301** notifies the base station device **101** (network) of information by which it is possible to specify at least one of a combination of antennas that have a predetermined characteristic in common and a combination of antennas that do not have the predetermined characteristic in common among the plurality of antennas included in the terminal device **111**, for example, as described above. Note that in addition to that information, the information notification unit **301** may notify the base station device **101** of information specifying the predetermined characteristic. For example, the information notification unit **301** can notify the base station device **101** of information designating a predetermined characteristic such as average delay, delay spread, and direction of arrival of electromagnetic waves, and information by which it is possible to specify at least one of a combination of antennas that have the predetermined characteristic in common and a combination of antennas that do not have the predetermined characteristic in common. Note that if information regarding one of the plurality of types of predetermined characteristics that is to be subjected to notification is determined in advance, the information notification unit **301** does not need to notify the base station device **101** of the information indicating the predetermined characteristic. The processing unit **302** receives instruction information for control and setting from the base station device **101** and executes setting processing and control processing based on the instruction information.

(25) FIG. 4 is a diagram illustrating an example of a functional configuration of the base station device **101**. The base station device **101** includes, for example, an information acquisition unit **401** and a setting control unit **402**. The functions illustrated in FIG. 4 can be realized, for example, by the processor **201** executing a program stored in the ROM **202** or the storage device **204**. Note that this is just one example, and some or all of the functions shown in FIG. 4 may be realized by dedicated hardware. In addition, FIG. 4 shows only the functions related to this embodiment among the functions included in the base station device **101**, and the base station device **101** can of course have functions other than those shown in FIG. 4, such as functions naturally included in a base station device in a cellular communication system.

(26) The information acquisition unit **401** acquires information from the terminal device **111** according to which it is possible to specify at least one of a combination of antennas that have the predetermined characteristic in common and a combination of antennas that do not have the predetermined characteristic in common among the plurality of antennas included in the terminal device **111**. In addition, if information regarding one of the plurality of types of predetermined characteristics that is to be subjected to notification is not determined in advance, the information acquisition unit **401** can acquire information designating information relating to one of the predetermined characteristics, according to which it is possible to specify at least one of a combination of antennas that have the predetermined characteristic in common and a combination of antennas that do not have the predetermined characteristic in common. The setting control unit **402** executes setting and control of the terminal device **111** based on the information acquired by the information acquisition unit **401**.

(27) Flow of Processing

(28) Next, an example of a flow of processing executed between the base station device **101** and the terminal device **111** will be described.

(29) FIG. 5 shows an example of a flow of processing relating to uplink transmission power control when the predetermined characteristic relates to an average delay. In this processing, first, for example, when the terminal device **111** is in a connected state, the base station device **101** transmits a message inquiring about capability information to the terminal device **111** (step S501). Then, in response to the message, the terminal device **111** notifies the base station device **101** of capability

information of the terminal device **111** (step S502). At this time, the terminal device **111** can notify the base station device **101** that the average delay characteristic when electromagnetic waves arrive from the base station device **101** is the same for the antennas **121** and **122**, and similarly, the average delay characteristic is the same for the antennas **123** and **124**. Note that if it is determined in advance that antennas will be classified based on the average delay characteristic, notification that the predetermined characteristic is a characteristic related to the average delay may be omitted. In this case, the terminal device **111** may notify the base station device **101** only of information indicating that the antenna **121** and the antenna **122** are an antenna group that has the predetermined characteristic in common, and that the antenna **123** and the antenna **124** are an antenna group that has the predetermined characteristic in common. In addition, the terminal device **111** may notify the base station device **101** that the antenna **121** (or the antenna **122**) and the antennas **123** and **124** do not have the predetermined characteristic in common.

(30) The base station device **101** executes setting and control of communication of the terminal device **111**, based on the information received in step S502. For example, the base station device **101** sets the transmission power control for the antennas **121** and **122** and sets the transmission power control for the antennas **123** and **124** separately. The base station device **101** transmits, for example, an RRC reconfiguration message including a transmission power setting for each antenna group to the terminal device **111** (step S503). In response to this, the terminal device **111** transmits an RRC reconfiguration complete message to the base station device **101** (step S504), and ends the setting. Note that a conventional terminal device cannot perform notification of information according to which it is possible to specify at least one of an antenna group that has the predetermined characteristic in common and an antenna group that does not have the predetermined characteristic in common, as in step S502. For this reason, the base station device **101** performs one common setting for the plurality of antennas for such a conventional terminal device, regardless of the arrangement or the like of the antennas. In contrast, in this embodiment, the base station device **101** can determine, based on information obtained from the terminal device **111**, whether or not a common setting can be used for each of the antennas included in the terminal device **111**, and can execute setting control based on the result of that determination.

(31) Thereafter, the base station device **101** allocates, for example, frequency and time resources for uplink signal transmission to the terminal device (step S505), and the terminal device **111** transmits uplink data (Physical Uplink Shared Channel, PUSCH) using each antenna based on the allocation (step S506). The base station device **101** determines whether transmission power control is necessary for each antenna included in the terminal device **111**, for example, by measuring a predetermined signal, such as a demodulation reference signal (DM-RS), among the signals transmitted from the antennas **121** to **124**. Here, it is assumed that the base station device **101** has determined that the received power of the DM-RS transmitted from the antenna **121** and the antenna **122** is low and that the transmission power is to be increased (step S507). In this case, the base station device **101** increases the transmission power of the antenna **121** and the antenna **122**, while transmitting an instruction to the terminal device **111** to keep the transmission power of the antenna **123** and the antenna **124** unchanged. For example, the base station device **101** can transmit such an instruction included in downlink control information (DCI) that allocates resources for uplink data transmission to the terminal device **111** (step S508). Note that, for example, information in which identification information allocated to each setting in step S503 (“A” or “B” in FIG. 5) is associated with a transmission power control command can be transmitted to the terminal device **111**. Upon receiving this instruction, the terminal device **111** increases the transmission power of the antenna **121** and the antenna **122**, and transmits uplink data with the allocated resources, using each antenna (step S509).

(32) When receiving this uplink data, the base station device **101** also monitors the DM-RS and continuously executes the transmission power according to the reception power. Here, it is assumed that the base station device **101** has determined that the reception power of the DM-RS transmitted

from the antenna **123** and the antenna **124** is high and that the transmission power is to be reduced (step **S510**). In this case, the base station device **101** transmits an instruction to the terminal device **111** to reduce the transmission power of the antenna **123** and the antenna **124**, while keeping the transmission power of the antenna **121** and the antenna **122** unchanged. For example, the base station device **101** can transmit such an instruction included in downlink control information (DCI) that allocates resources for uplink data transmission to the terminal device **111** (step **S511**). Then, upon receiving this instruction, the terminal device **111** reduces the transmission power of the antenna **123** and the antenna **124**, and transmits uplink data with the allocated resources, using each antenna (step **S512**).

(33) Note that although FIG. 5 shows an example in which the increase in transmission power of the antenna **121** and the antenna **122** and the decrease in transmission power of the antenna **123** and the antenna **124** are performed at different timings, these controls may be performed simultaneously. For example, the base station device **101** can transmit, to the terminal device **111**, a DCI including an instruction to increase the transmission power of the antenna **121** and the antenna **122** and decrease the transmission power of the antenna **123** and the antenna **124**. Note that it is sufficient that the antennas **121** and **122** and the antennas **123** and **124** are controlled independently, and instructions may be given to increase or decrease the transmission power for all of these antennas. However, in this case as well, the instruction can be transmitted for each antenna group having a predetermined characteristic in common.

(34) The above-mentioned processing makes it possible to appropriately control the transmission power for each antenna, and to improve the efficiency of communication.

(35) FIG. 6 shows an example of a flow of setting control processing related to reporting of a beam in the case where the predetermined characteristic relates to the direction in which electromagnetic waves from the base station device **101** arrive at the terminal device **111**. In this processing, similarly to the case of FIG. 5, first, the base station device **101** transmits a message inquiring about capability information to the terminal device **111** (step **S601**). Then, in response to the message, the terminal device **111** notifies the base station device **101** of the capability information of the terminal device **111** (step **S602**). At this time, the terminal device **111** can notify the base station device **101** that the antennas **121** and **122** have the characteristic of the direction (Angle of Arrival, AoA) in which electromagnetic waves arrive from the base station device **101** in common, and that the antennas **123** and **124** similarly have the AoA characteristic in common. Note that if it is determined in advance that antenna classification will be performed based on the AoA characteristic, notification that the predetermined characteristic is a characteristic related to AoA may be omitted.

(36) The base station device **101** executes setting and control of communication of the terminal device **111**, based on the information received in step **S602**. The base station device **101** transmits, for example, an RRC reconfiguration message including settings for beam control and management for each antenna group to the terminal device **111** (step **S603**). In response to this, the terminal device **111** transmits an RRC reconfiguration complete message to the base station device **101** (step **S604**), and ends the setting. The base station device **101** notifies the terminal device **111** of a channel state information-reference signal (CSI-RS) setting and a reporting setting therefor, for example, as settings necessary for beam control and management, for each antenna group that has the predetermined characteristic in common. Note that in this processing, for conventional terminal devices, one common setting is made for a plurality of antennas regardless of their arrangement. In contrast, in this embodiment, the base station device **101** can specify, based on information obtained from the terminal device **111**, whether or not a common setting can be used for each of the antennas included in the terminal device **111**, and can execute setting control based on the result of that specification.

(37) Thereafter, the base station device **101** transmits, for example, a CSI-RS corresponding to the settings for the antennas **121** and **122** (step **S605**), and the terminal device **111** measures the CSI-

RS using the antennas **121** and **122** (step **S606**). Note that the terminal device **111** can be configured not to perform measurement using the antennas **123** and **124** for this CSI-RS. In addition, the base station device **101** transmits, for example, a CSI-RS corresponding to the settings for the antennas **123** and **124** (step **S607**), and the terminal device **111** measures the CSI-RS using the antennas **123** and **124** (step **S608**). Note that the terminal device **111** can be configured not to perform measurement using the antennas **121** and **122** for this CSI-RS. The terminal device **111** continuously executes CSI-RS measurement based on a first setting relating to the antennas **121** and **122**, using the antennas **121** and **122** (step **S609**). Then, if the measurement result satisfies a condition designated by a report setting relating to the antennas **121** and **122** (step **S610**), the terminal device **111** notifies the base station device **101** of the measurement result (step **S611**). Similarly, the terminal device **111** continuously executes CSI-RS measurement based on a second setting relating to the antennas **123** and **124**, using the antennas **123** and **124** (step **S612**). Then, if the measurement result satisfies a condition designated by the report setting for the antennas **123** and **124** (step **S613**), the terminal device **111** notifies the base station device **101** of the measurement result (step **S614**).

(38) Based on the notified measurement result, the base station device **101** performs beam control and management in the conventional manner. By performing such processing, it is possible to appropriately control and manage beams for each antenna, thereby improving communication efficiency.

(39) FIG. 7 shows an example of a flow of setting control processing for a sounding reference signal in a case where the predetermined characteristic relates to delay spread. In this processing as well, similarly to the cases of FIGS. 5 and 6, first, the base station device **101** transmits a message inquiring about capability information to the terminal device **111** (step **S701**). Then, in response to the message, the terminal device **111** notifies the base station device **101** of the capability information of the terminal device **111** (step **S702**). At this time, the terminal device **111** can notify the base station device **101** that the delay spread characteristics when the electromagnetic waves transmitted from the antennas **121** and **122** arrive at the base station device **101** are the same, and that the delay spread characteristics of the antennas **123** and **124** are the same. Note that if it is determined in advance that antenna classification will be performed based on the delay spread characteristics, notification that the predetermined characteristic is a characteristic relating to AoA may be omitted.

(40) The base station device **101** executes setting and control of communication of the terminal device **111**, based on the information received in step **S702**. The base station device **101** transmits, for example, an RRC reconfiguration message including a setting of a sounding reference signal (SRS) for each antenna group to the terminal device **111** (step **S703**). In response to this, the terminal device **111** transmits an RRC reconfiguration complete message to the base station device **101** (step **S704**), and ends the setting. The base station device **101** can set, for example, the subcarrier spacing for transmitting the SRS (the density of the SRS on the frequency axis) separately for the antennas **121** and **122** and the antennas **123** and **124**. Thereafter, the terminal device **111** uses the antennas **121** and **122** to transmit the SRS based on a first SRS setting for those antennas, and uses the antennas **123** and **124** to transmit the SRS based on a second SRS setting for those antennas. This makes it possible to suppress the deterioration of efficiency caused by using a common setting for antennas for which the states of the transmission paths are expected to differ greatly. Note that in this processing as well, for a conventional terminal device, one common setting is made for a plurality of antennas regardless of their arrangement. In contrast, in this embodiment, the base station device **101** can specify, based on information obtained from the terminal device **111**, whether or not a common setting can be used for the antennas included in the terminal device **111**, and can execute setting control based on the result of that specification.

(41) Note that the processing shown in FIGS. 5 to 7 is merely an example, and processing other than the setting and control processing shown may also be executed. Also, in FIGS. 5 to 7, it has

been described that the antennas are classified according to the average delay, AoA, and delay spread, but in reality, these can be decided on by only the arrangement of the antennas. That is, antennas whose inter-antenna distance is within a predetermined range can be treated as a group having a characteristic in common, and antennas whose inter-antenna distance exceeds the predetermined range can be treated as having different characteristics. In addition, the distance and positional relationship of the antenna arrangement may be associated in advance with whether or not there is commonality of a predetermined characteristic, and based on the characteristic used for setting control, the terminal device may notify the network of at least one of an antenna group for which common setting control is to be performed and an antenna group for which independent setting control is to be performed. For example, in the above example, it is assumed that the antennas **121** and **122** have a characteristic in common, and the antennas **123** and **124** have characteristics different from the antennas **121** and **122**, but have a characteristic in common between the antennas **123** and **124**, but there is no limitation to this. For example, with regard to the average delay, the antennas **121** and **122** may have a characteristic in common, and the antennas **123** and **124** may have a characteristic in common, while with regard to the delay spread, the antennas **121** and **124** may have a characteristic in common, and the antennas **122** and **123** may have a characteristic in common, or the like. In this manner, the plurality of antennas are classified into antenna groups each including one or more antennas, but the classification may vary according to a predetermined characteristic.

(42) In this manner, it becomes possible to perform appropriate communication control in response to the diversification of antenna arrangements in terminal devices. This enables contribution to Goal 9 of the United Nations Sustainable Development Goals (SDGs), which is to “build resilient infrastructure, promote sustainable industrialization and foster innovation”.

(43) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

Claims

1. A terminal device comprising: a plurality of antennas; at least one processor; and at least one memory that stores a computer-readable instruction for causing, when executed by the at least one processor, the at least one processor to: notify a network to which the terminal device belongs of information that identifies at least one of, among the plurality of antennas, a second antenna having a predetermined characteristic in common with a first antenna included in the plurality of antennas, and a third antenna not having the predetermined characteristic in common with the first antenna.
2. The terminal device according to claim 1, wherein the terminal device notifies the network of the predetermined characteristic as well as the information.
3. The terminal device according to claim 1, wherein the predetermined characteristic relates to a magnitude of delay observed regarding electromagnetic waves that arrive from a base station device belonging to the network or a magnitude of delay obtained regarding electromagnetic waves that arrive at the base station device from the terminal device.
4. The terminal device according to claim 1, wherein the predetermined characteristic relates to a magnitude of delay spread observed regarding electromagnetic waves that are transmitted from the terminal device and arrive at a base station device belonging to the network.
5. The terminal device according to claim 1, wherein the predetermined characteristic relates to a direction in which electromagnetic waves arrive from a base station device belonging to the network.
6. The terminal device according to claim 1, wherein in response to a request for information on a capability of the terminal device from the network, the terminal device performs notification of the

information.

7. A network node comprising: at least one processor; and at least one memory that stores a computer-readable instruction for causing, when executed by the at least one processor, the at least one processor to: acquire, from a terminal device belonging to a network including the network node, information that identifies at least one of, among a plurality of antennas included in the terminal device, a second antenna having a predetermined characteristic in common with a first antenna included in the plurality of antennas, and a third antenna not having the predetermined characteristic in common with the first antenna; and perform, based on the information, control of communication of the terminal device, the control being the same for the first antenna and the second antenna, and the control being different for the first antenna and the third antenna.
 8. The network node according to claim 7, wherein the network node acquires the predetermined characteristic as well as the information from the terminal device.
 9. The network node according to claim 7, wherein the predetermined characteristic relates to a magnitude of delay observed regarding electromagnetic waves that arrive from a base station device belonging to the network or a magnitude of delay obtained regarding electromagnetic waves that arrive at the base station device from the terminal device, and the network node performs common transmission power control for the first antenna and the second antenna and performs independent transmission power control for the first antenna and the third antenna.
 10. The network node according to claim 7, wherein the predetermined characteristic relates to a magnitude of delay of electromagnetic waves arriving at the terminal device from a base station device belonging to the network, and the network node performs common transmission timing control for the first antenna and the second antenna and performs independent transmission timing control for the first antenna and the third antenna.
 11. The network node according to claim 7, wherein the predetermined characteristic relates to a delay spread observed regarding electromagnetic waves that are transmitted from the terminal device and arrive at a base station device belonging to the network, and the network node performs common sounding reference signal setting for the first antenna and the second antenna and performs independent sounding reference signal setting for the first antenna and the third antenna.
 12. The network node according to claim 7, wherein the predetermined characteristic relates to a direction in which electromagnetic waves transmitted from a base station device belonging to the network arrive at the terminal device, and the network node performs common beam control for the first antenna and the second antenna and performs independent beam control for the first antenna and the third antenna.
 13. The network node according to claim 7, wherein the network node transmits a request for information on a capability to the terminal device, and acquires the information as a response to the request.
 14. A control method to be executed by a terminal device including a plurality of antennas, comprising: notifying a network to which the terminal device belongs of information that identifies at least one of, among the plurality of antennas, a second antenna having a predetermined characteristic in common with a first antenna included in the plurality of antennas, and a third antenna not having the predetermined characteristic in common with the first antenna.
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